DE LA CRUZ GABRIEL

Senior Unreal Engine Programmer

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Portfolio: gabriel98pc.github.io/GabrielDeLaCruz.github.io/

PROFESSIONAL SUMMARY

Results-driven Senior Unreal Engine Developer with over 4 years of experience architecting interactive solutions, multiplayer frameworks, and custom Engine plugins across shipped Steam titles and VR experiences. Specialized in C++ and Blueprints with a rigorous focus on performance optimization, procedural animation, and scalable systems architecture. Proven track record of mitigating CPU bottlenecks in complex scenes and integrating robust backend systems utilizing AWS.

TECHNICAL SKILLS

Programming Languages: C++, C#, Python, Java, Node.js, PHP.

Engine Expertise: Unreal Engine (C++ API, Blueprints, Networking/Multiplayer, AI, Custom Plugin Development, Engine Migration, MetaHumans), Unity (WebGL, UI Canvas, Android).

Key Competencies: Performance Optimization (Tick reduction, vector-array sampling), Procedural Locomotion Systems, Backend Integration.

Tools & Cloud: AWS API Gateway, Amazon GameLift, Git, Perforce, Blender, Adobe Photoshop, Adobe Premiere.

AI: Claude, Gemini, Comfy UI

PROFESSIONAL EXPERIENCE

Unreal Engine Programmer | Realidad360

2026 - Present

- Develop Unreal Engine plugins to complement the development of virtual reality experiences.
- Develop AI virtual assistants that guide users through the full experience, for both sighted and blind users, including a watchtower/guide system, mesh dissolve components, editor level-navigation tools, and a custom state machine tool for level behavior.
- Implement MetaHumans and build easily replicated systems in Blueprints for artist- and designer-facing tools.

Gameplay Programmer | RolldBox

2023 - 2026

- Engineered robust gameplay mechanics and developed custom Unreal Engine plugins utilizing C++ and Blueprints.
- Spearheaded critical performance optimization initiatives, successfully migrating heavy Tick-based calculations to pre-calculated spline data using vector-array sampling to eliminate CPU bottlenecks in entity-heavy scenes.
- Architected and integrated scalable backend systems for multiplayer games using AWS API Gateway.
- Managed seamless project environment upgrades between major engine versions (e.g., resolving serialization and crash issues migrating from UE 5.4 to 5.5).
- Shipped Mask of the Void (Steam), building its Ability System, Status Effect System, AI Behavior and custom AI Pathfinding systems, Interaction System, Player Progression System, multiplatform login, and user metrics system.
- Shipped Those Who Came: Healing Solarus (Steam), building its multiplatform login and AWS backend systems.

Unity/Unreal Developer | Lightbox Studios

2023 - 2026

- Developed high-fidelity interactive 3D solutions and configurators for PC and Web environments using Unreal Engine and Unity WebGL.
- Designed and implemented advanced procedural locomotion systems for complex skeletons, including multi-limbed (8-legged) and bipedal (raptor) characters, ensuring high-performance real-time animation.

Unreal Developer | Aguascalien3D

2023

- Programmed core gameplay systems and interactive logic leveraging a combination of C++ and Blueprint visual scripting to meet strict project milestones.

Freelance Unreal Developer | Upwork

2022 - 2023

- Delivered end-to-end development for multiplayer video games, handling everything from client-side prediction to server-side authority using Unreal Engine's networking framework.

Unity Developer | TGL

2022 - 2023

- Developed expansive metaverse environments in Unity, utilizing C# and Java to build stable, concurrent multiplayer solutions.

Unreal Developer | Indie Films

2021 - 2022

- Created real-time virtual production environments and visual solutions for television broadcasts utilizing Unreal Engine.
- Built a C++ web-request (API) data-driven plugin for live production data.

PUBLISHED PLUGINS & ASSETS

Developed and published proprietary tools demonstrating deep knowledge of the Unreal Engine source code and architecture:

- Anim Blend Tools Pro (Fab Store) - 10 animation baking tools, including Bone Blend & Layer Bakers, Control Rig Baker, Ragdoll Recorder, Root Motion Baker, and Speed Ramp Baker
- Cross Flow UI (Fab Store)
- Seamless Lobby Solution (Fab Store)
- Tickable Latent Objects (Fab Store)
- Spline Ribbon (Fab Store)
- Actor Function Trigger (Fab Store)

EDUCATION

- Object-Oriented Programming (OOP) Course | EXO Training Center
- Bachelor's Degree in Economics | EEM n°8 Julio Cortázar

LANGUAGES

- English: Intermediate (Technical proficiency)
- Spanish: Native

REFERENCES

Carlos López - https://www.linkedin.com/in/carlos3dgen/

Bosco Barber - https://www.linkedin.com/in/boscobarber/

Luis Merino - https://www.linkedin.com/in/luis-merino-san-roman-a116a41a9/

Ferran Pérez - https://www.linkedin.com/in/ferranperezperez/